Data modalities

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An uncomfortable mind search for meaning , happiness , freedom and philosophy of life .

A comfortable mind just see the world as it is.
(Nagesh)

Contents

Types of Data in Machine Learning 3

1. Grid Data 3
Mathematical Representation 4
Properties 4
Models Used 4
Examples 4

2. Sequential Data 5
Mathematical Representation 5
Properties 5
Models Used 5
Examples 6

3. Graph Data 6
Mathematical Representation 6
Properties 6
Models Used 7
Message Passing 7
Examples 7

4. Point Cloud / Set Data 7
Mathematical Representation 7
Important Property 8
Models Used 8
Deep Sets Formulation 8
Examples 8

5. Manifold Data **8**
Mathematical Idea 9
Examples 9
Models Used 9
Manifold Learning Methods 9
Example 10

6. Tree Data **10**
Mathematical Representation 10
Recursive Representation 10
Models Used 10
Examples 10

7. Spatio-Temporal Data **11**
Mathematical Representation 11
Properties 11
Models Used 11
Examples 12

8. Mesh / Surface Data **12**
Mathematical Representation 12
Properties 12
Models Used 13
Examples 13

Relationship Between Data Geometry and Models **13**

Unified View **13**

Key Insight **14**

Types of Data in Machine Learning

Different machine learning problems involve different underlying data structures.

The geometry and topology of data strongly influence:

- representation learning
- model architecture
- inductive biases

A major theme in deep learning is:

$$\boxed{\text{Architecture should respect data structure}}$$

Common data types include:

1. Grid Data
2. Sequential Data
3. Graph Data
4. Point Cloud / Set Data
5. Manifold Data
6. Tree Data
7. Spatio-Temporal Data
8. Mesh / Surface Data

1. Grid Data

Grid data has regular Euclidean structure.

Examples:

- images
- videos
- volumetric medical scans

Mathematical Representation

2D image:

$$X \in \mathbb{R}^{H \times W \times C}$$

where:

- H = height
- W = width
- C = channels

Example:
RGB image:

$$C = 3$$

Properties

- regular lattice structure
- local spatial correlations
- translation symmetry

Models Used

- CNNs
- Vision Transformers (ViTs)
- UNets
- Diffusion models

Examples

- MNIST
- CIFAR-10
- MRI scans
- satellite imagery

2. Sequential Data

Sequential data has temporal or ordered structure.

Examples:

- text
- speech
- DNA sequences
- financial time series

Mathematical Representation

Sequence:

$$X = \{x_1, x_2, \ldots, x_T\}$$

where:

$$x_t \in \mathbb{R}^d$$

Thus:

$$X \in \mathbb{R}^{T \times d}$$

Properties

- ordering matters
- causal dependencies
- temporal correlations

Models Used

- RNNs
- LSTMs
- GRUs
- Transformers
- State-space models

Examples

- natural language
- ECG signals
- protein sequences
- stock market data

3. Graph Data

Graph data represents relational structures.

A graph:

$$G = (V, E)$$

where:

- V = nodes
- E = edges

Mathematical Representation

Adjacency matrix:

$$A \in \mathbb{R}^{N \times N}$$

where:

$$A_{ij} = 1$$

if edge exists.

Node features:

$$X \in \mathbb{R}^{N \times d}$$

Properties

- irregular connectivity
- non-Euclidean structure
- relational interactions

Models Used

- Graph Neural Networks (GNNs)
- Graph Convolutional Networks (GCNs)
- Graph Attention Networks (GATs)
- Message Passing Networks

Message Passing

Typical GNN update:

$$h_i^{(k+1)} = \phi\left(h_i^{(k)}, \sum_{j \in \mathcal{N}(i)} \psi(h_i^{(k)}, h_j^{(k)})\right)$$

Examples

- social networks
- molecular graphs
- citation networks
- transportation networks

4. Point Cloud / Set Data

Point clouds are unordered collections of points.

Examples:

- LiDAR scans
- 3D objects
- particle simulations

Mathematical Representation

Point set:

$$X = \{x_1, x_2, \ldots, x_N\}$$

where:

$$x_i \in \mathbb{R}^d$$

Example:

3D point cloud:

$$x_i = (x, y, z)$$

Important Property

Sets are permutation invariant.

Meaning:

$$\{x_1, x_2\} = \{x_2, x_1\}$$

Ordering should not affect output.

Models Used

- PointNet
- PointNet++
- Set Transformers
- Deep Sets

Deep Sets Formulation

Permutation invariant mapping:

$$f(X) = \rho\left(\sum_{i=1}^{N} \phi(x_i)\right)$$

Examples

- autonomous driving LiDAR
- protein atomic coordinates
- galaxy distributions

5. Manifold Data

Many high-dimensional datasets lie near low-dimensional manifolds.

Mathematical Idea

Suppose:

$$x \in \mathbb{R}^D$$

but intrinsic dimension:

$$d \ll D$$

Then data lies on manifold:

$$\mathcal{M} \subset \mathbb{R}^D$$

with:

$$\dim(\mathcal{M}) = d$$

Examples

- natural images
- speech signals
- molecular conformations

Models Used

- Autoencoders
- Variational Autoencoders
- Diffusion models
- Manifold learning algorithms

Manifold Learning Methods

- PCA
- t-SNE
- UMAP
- Isomap
- Laplacian Eigenmaps

Example

Human face images may exist in:

$$\mathbb{R}^{100000}$$

pixel space, but intrinsic manifold dimension may be much smaller.

6. Tree Data

Tree data contains hierarchical structure.
Examples:

- syntax trees
- phylogenetic trees
- file systems

Mathematical Representation

A tree:

$$T = (V, E)$$

with no cycles.
Hierarchical parent-child relationships.

Recursive Representation

Node representation:

$$h_i = f\left(x_i, \{h_j : j \in \text{children}(i)\}\right)$$

Models Used

- Recursive Neural Networks
- Tree-LSTMs
- Hierarchical Transformers

Examples

- sentence parse trees
- evolutionary trees
- XML/HTML documents

7. Spatio-Temporal Data

Spatio-temporal data evolves across both space and time.

Examples:

- videos
- climate systems
- traffic flow
- fluid dynamics

Mathematical Representation

Suppose:

$$X(t, s)$$

where:

- t = time
- s = spatial coordinate

Example:

$$X \in \mathbb{R}^{T \times H \times W \times C}$$

for videos.

Properties

- spatial correlations
- temporal evolution
- dynamical behavior

Models Used

- ConvLSTM
- Video Transformers
- Neural PDE models
- Spatio-temporal GNNs

Examples

- weather prediction
- surveillance video
- ocean currents
- reaction-diffusion systems

8. Mesh / Surface Data

Meshes represent geometric surfaces.
Widely used in:

- computer graphics
- finite element methods
- molecular geometry

Mathematical Representation

Mesh:

$$M = (V, E, F)$$

where:

- V = vertices
- E = edges
- F = faces

Example:
triangle mesh:

$$F = (v_i, v_j, v_k)$$

Properties

- irregular geometry
- local neighborhoods
- surface topology

Models Used

- MeshCNN
- Geometric Deep Learning
- Graph Neural Networks
- Neural Operators

Examples

- 3D CAD models
- protein surfaces
- finite element meshes
- human body scans

Relationship Between Data Geometry and Models

Different architectures encode different symmetries.

Data Type	Main Symmetry / Structure
Grid	Translation symmetry
Sequential	Temporal causality
Graph	Relational topology
Point Cloud	Permutation invariance
Manifold	Low-dimensional geometry
Tree	Hierarchical structure
Spatio-temporal	Space-time dynamics
Mesh	Surface topology

Unified View

Modern machine learning architectures are fundamentally attempts to learn:

$$\boxed{\text{Transformations respecting data geometry}}$$

CNNs exploit:

$$\text{Euclidean grids}$$

Transformers exploit:

$$\text{sequence interactions}$$

GNNs exploit:

$$\text{graph topology}$$

Neural operators exploit:

$$\text{function space dynamics}$$

Thus data geometry determines model architecture.

Key Insight

Machine learning architectures are deeply connected to:

- topology
- geometry
- symmetry
- structure of data

The success of deep learning largely comes from designing models that respect the intrinsic structure of data manifolds.